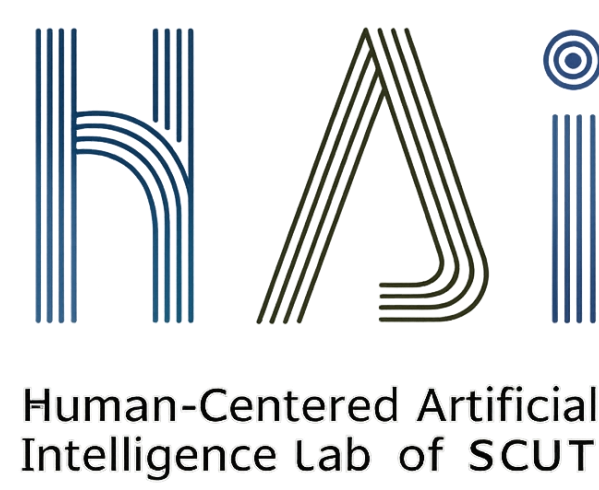


Persona-E²: A Human-Grounded Dataset for Personality-Shaped Emotional Responses to Textual Events



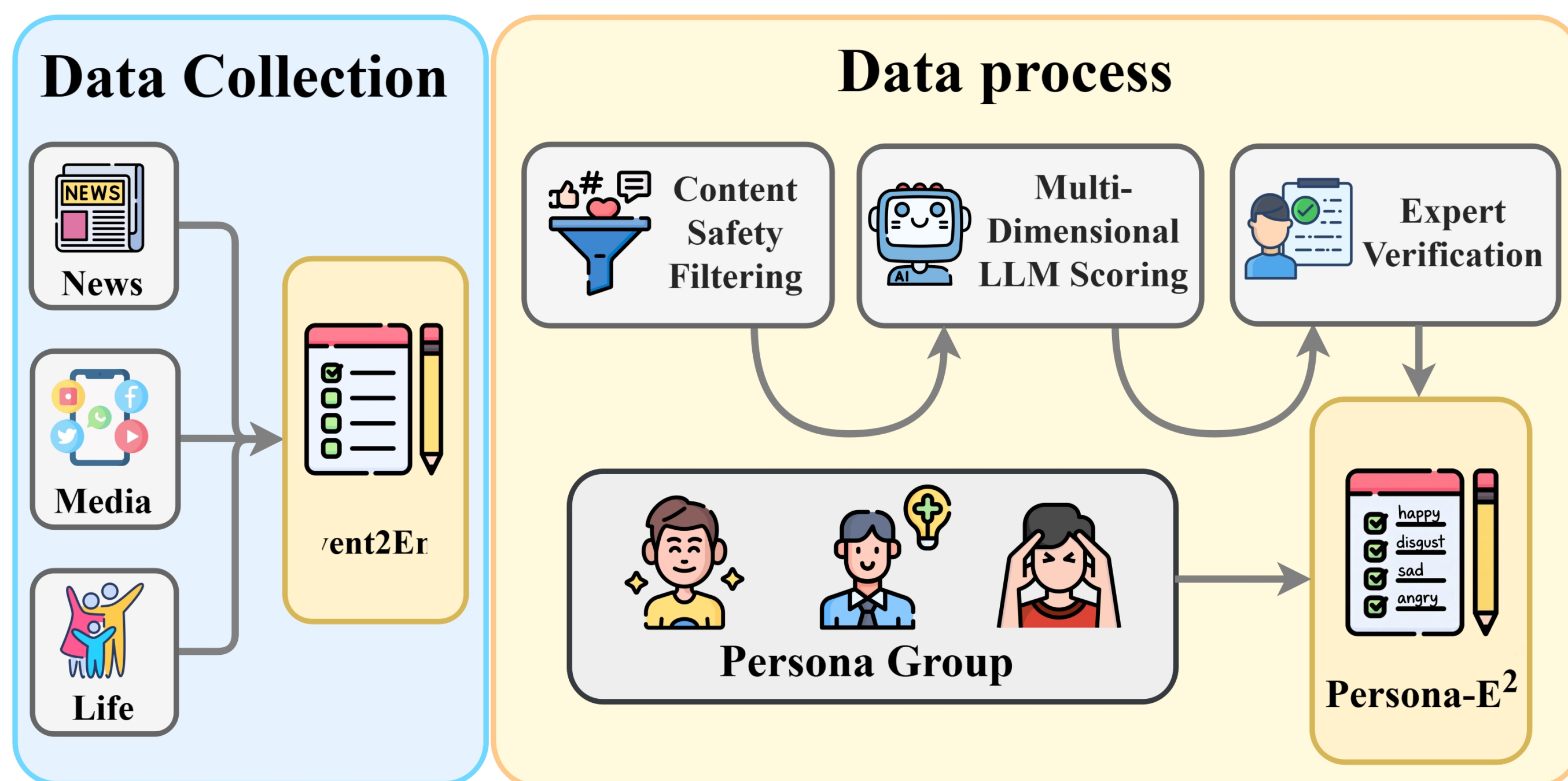
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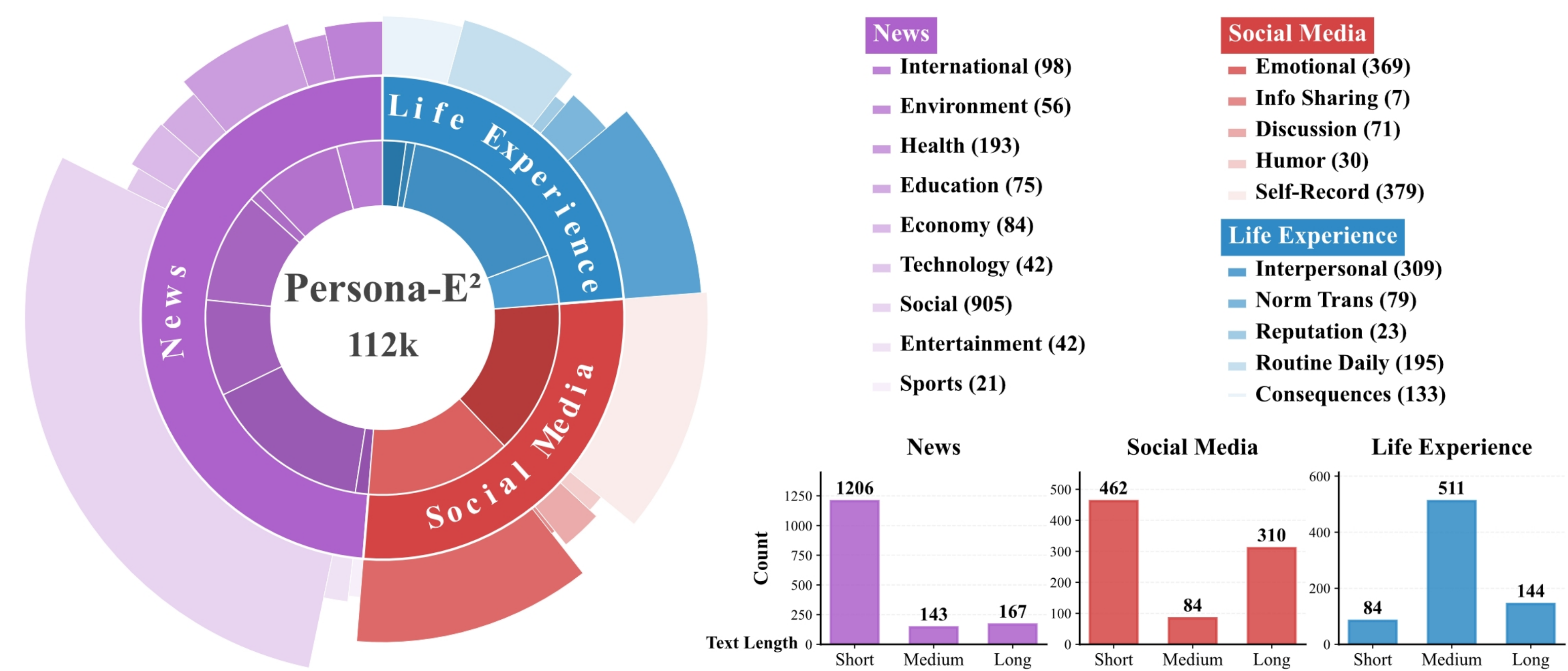
Persona-E²

- Current affective computing overlooks personality-driven reader emotions, while LLMs suffer from personality illusion. We introduce Persona-E², a human-annotated dataset linking MBTI/Big Five traits to reader emotional responses across domains.

Data Construction Pipeline



Data Source



Experiments

Research Questions

RQ1

Affective Divergence

How do emotions vary across writers, readers, and personalities across domains?

RQ2

LLM Simulation

Can LLMs simulate personality-driven emotional responses accurately?

RQ3

Cognitive Soundness

Do LLMs produce psychologically valid emotion reasoning with personality?

Models Evaluated



Top-K emotion distribution

Groups	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Overall
	58.9	29.4	11.7	0.0	0.0	0.0	0.0
	56.0	26.3	14.9	2.8	0.0	0.0	0.0
	56.8	24.7	12.2	5.0	1.2	0.1	0.0
	49.0	24.6	14.2	7.8	3.5	0.8	0.1
	55.1	25.5	12.8	5.6	1.0	0.0	0.0
	54.5	25.9	13.9	5.1	0.6	0.0	0.0
	47.4	23.5	14.0	8.3	4.4	1.9	0.5
	Top1	Top2	Top3	Top4	Top5	Top6	Top7

- Fig.4 shows the top-K emotion vote share across BFI personality annotation clusters, revealing a dominant emotion across groups and providing a human response baseline for LLM simulation.

Findings

Finding 1: Emotional Divergence (Writer vs Reader vs Persona Reader)

- Emotion is constructed, not transferred.

	News			Social Media			Life Experience		
W/R	Pos	Neu	Neg	W/R	Pos	Neu	W/R	Pos	Neu
Pos	48.10	26.30	25.60	22.25	18.40	59.35	62.40	5.55	32.05
Neu	28.72	43.66	27.62	4.70	13.70	81.60	56.20	13.00	30.80
Neg	21.07	22.33	56.60	11.78	15.35	72.87	43.28	6.65	50.07

Domain effects:

- News**: Neutral & Stable, lowest divergence.
- Social**: Negativity bias, "Emotional Black Hole".
- Life**: Positive shift, "Psychological Immune system".

Finding 2: Personality Matters

- Personality traits shape emotional transitions.
- $PAG = Agr_{in} - Agr_{out}$
- Agr_{in} : Top-1 agreement within the same personality cluster
- Agr_{out} : Top-1 agreement between different personality clusters
- Examples (Spearman ρ with valence change):

Grouping Method	PAG mean	PAG max	PAG min
BFI K-means ($K = 6$)	16.69	25.96	8.27
BFI K-means ($K = 4$)	13.13	20.42	5.90
MBTI Grouping ($N \geq 3$)	20.15	26.98	9.68

$PAG > 0$ across domains $\rightarrow$ structured disagreement (not noise).

Finding 3: LLM Limitations

- LLMs capture emotion neighborhood but struggle with precise prediction.

Source	News		Social Media		Life Experience		Overall	
Prompt	General	CoT	General	CoT	General	CoT	General	CoT
GPT5.1	31.8	25.0	29.5	18.2	18.2	27.3	32.4	35.3
LLAMA3-8B	29.5	29.3	20.5	9.1	4.8	13.6	32.4	32.4
QWEN3-8B	18.2	26.9	25.4	23.1	21.4	22.7	20.3	33.7
GEMMA3-12B	18.2	25.0	27.3	22.7	27.3	18.2	35.3	32.4
MINISTRAL3-8B	18.2	13.6	20.5	5.0	9.1	4.5	36.4	35.3

Table 3: Comparison of different models performance in subset. Note: General Prompt: Prompt without personality traits, Persona: Prompt with BFI traits, CoT: CoT-Style Prompt with BFI traits.

LLMs find the neighborhood, but miss subtle emotion shifts. Social media is the hardest domain.

Finding 4: BFI > MBTI

- Personality information helps, and BFI outperforms MBTI consistently.

Model	Consistency			Plausibility			Specificity		
	BL	MBTI	BFI	BL	MBTI	BFI	BL	MBTI	BFI
GPT5.1	4.7	16.5	78.8	13.8	15.8	70.4	17.6	13.5	68.9
LLAMA3	9.4	19.8	70.8	21.5	23.1	55.4	20.5	18.2	61.3
QWEN3	10.4	19.4	70.2	19.7	20.4	59.9	23.7	19.4	56.9
GEMMA3	14.8	17.4	67.8	21.8	15.1	63.1	19.5	15.5	65.0
MINISTRAL3	12.5	15.5	72.0	21.6	19.5	58.9	22.0	12.5	65.5

Table 4: Win-Rate comparison of LLMs in the best-of-3 selection task. Note: BL: Baseline Prompt without personality, MBTI: MBTI Prompt, BFI: BFI Prompt.

Takeaways

- Emotion is not a property of text, but of the reader.
- Personality is a core latent variable in emotion elicitation.
- Social media is the most challenging domain for both humans and LLMs.
- Future NLP needs persona-aware modeling and evaluation.



Project Page

<https://scut-hai.github.io/Persona-E2/>



Dataset

Kaggle:
Persona-E²: Personality-Shaped Emotion Dataset
Huggingface:
Persona-E2-Dataset



Contact

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Scan for more

Huggingface



Kaggle



Homepage

